Domain-Aware Machine Learning

Bernhard Geiger (TU Graz, KNOW) & Franz Rohrhofer (KNOW)

NextAIRE

Version: May 7, 2026

Acknowledgments

Based on Course 442.026 @ Graz University of Technology

Hybrid Modeling
and
Domain-Aware Machine Learning

- Introducing approaches to domain-aware ML *and* hybrid modeling
- In-depth discussion of selected approaches and applications
- Guest talks from experts
- Tuesday, IDEG134, 10:00-12:00
- 2 VU $\rightarrow$ reading group presentation, (small) homework, contribution to course notes

Definitions

Domain-Aware ML

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Lecture Timeline

1 Definitions and Motivation

2 A Brief Intro to Machine Learning

3 Hybrid Modeling and Domain-Aware Machine Learning – Overview

What is Machine Learning?

What is Machine Learning?

Machine learning is a discipline focused on two interrelated questions: **How can one construct** *computer systems that automatically* **improve through experience***? and What are the* **fundamental** *statistical-computational-information-theoretic* **laws** *that govern all learning systems, including computers, humans, and organizations? [...] A learning problem can be defined as the problem of improving some measure of performance when executing some* **task***, through some type of training experience.*

—M. I. Jordan and T. M. Mitchell. "Machine learning: trends, perspectives, and prospects". Science 349.6245 (July 2015), pp. 255–260.

What is Machine Learning?

Machine learning (ML) is a branch of artificial intelligence concerned with the development of algorithms that improve their performance when more information becomes available, typically in the form of **data**. *At its core, ML seeks to* **approximate functions or distributions** *based on observed input-output pairs, often* **without explicit knowledge** *of the underlying generative process.*

—F. Dietrich and W. Schilders. "Scientific machine learning". Math Semesterber (2025).

What is Machine Learning?

Machine learning (ML) is a field of study in artificial intelligence concerned with the **development and study** *of statistical algorithms that can learn from* **data** *and generalise to unseen data, and thus perform* **tasks** *without explicit instructions.*

—Wikipedia, accessed 17.09.2025

What is Domain-Aware ML?

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What is needed is an approach that leverages the advances in data-driven research yet **constrains both the methods and the interpretation of the results** *through tried-and-true scientific theory. Thus, to make significant contributions [...], new data science* **methods must encapsulate domain knowledge** *to produce theoretically-consistent results.*

—J. H. Faghmous and V. Kumar. "A Big Data Guide to Understanding Climate Change: The Case for Theory-Guided Data Science". Big Data 2:3 (2014), pp. 155-163

What is Domain-Aware ML?

[T]here is a growing interest [...] on creating a new generation of methods that **integrate scientific knowledge in ML** *frameworks. This emerging field, called scientific knowledge-guided ML (KGML), seeks a distinct departure from existing "data-only" or "scientific knowledge-only" methods to use knowledge and data at an equal footing.*

—Opening of A. Karpatne, R. Kannan, and V. Kumar. "Knowledge-Guided Machine Learning. Accelerating Discovery Using Scientific Knowledge and Data". CRC Press (2023)

What is Domain-Aware ML?

[...] need for integrating fundamental physical laws and domain knowledge by 'teaching' ML models about governing physical rules, which can, in turn, provide **'informative priors'** *— that is, strong theoretical constraints and inductive biases on top of the observational ones. [...] physics-informed learning is [...] the process by which* **prior knowledge** *stemming from our observational, empirical, physical or mathematical understanding of the world can be leveraged to* **improve the performance of a learning algorithm**.

—G.E. Karniadakis et al. "Physics-informed machine learning". Nat Rev Phys 3, 422–440 (2021)

What is Domain-Aware ML?

Definition

Domain-aware machine learning focuses on **incorporating domain knowledge** into machine learning. It seeks to approximate functions or distributions based on **observed input-output pairs** *and* an **understanding of the underlying data-generating process**.

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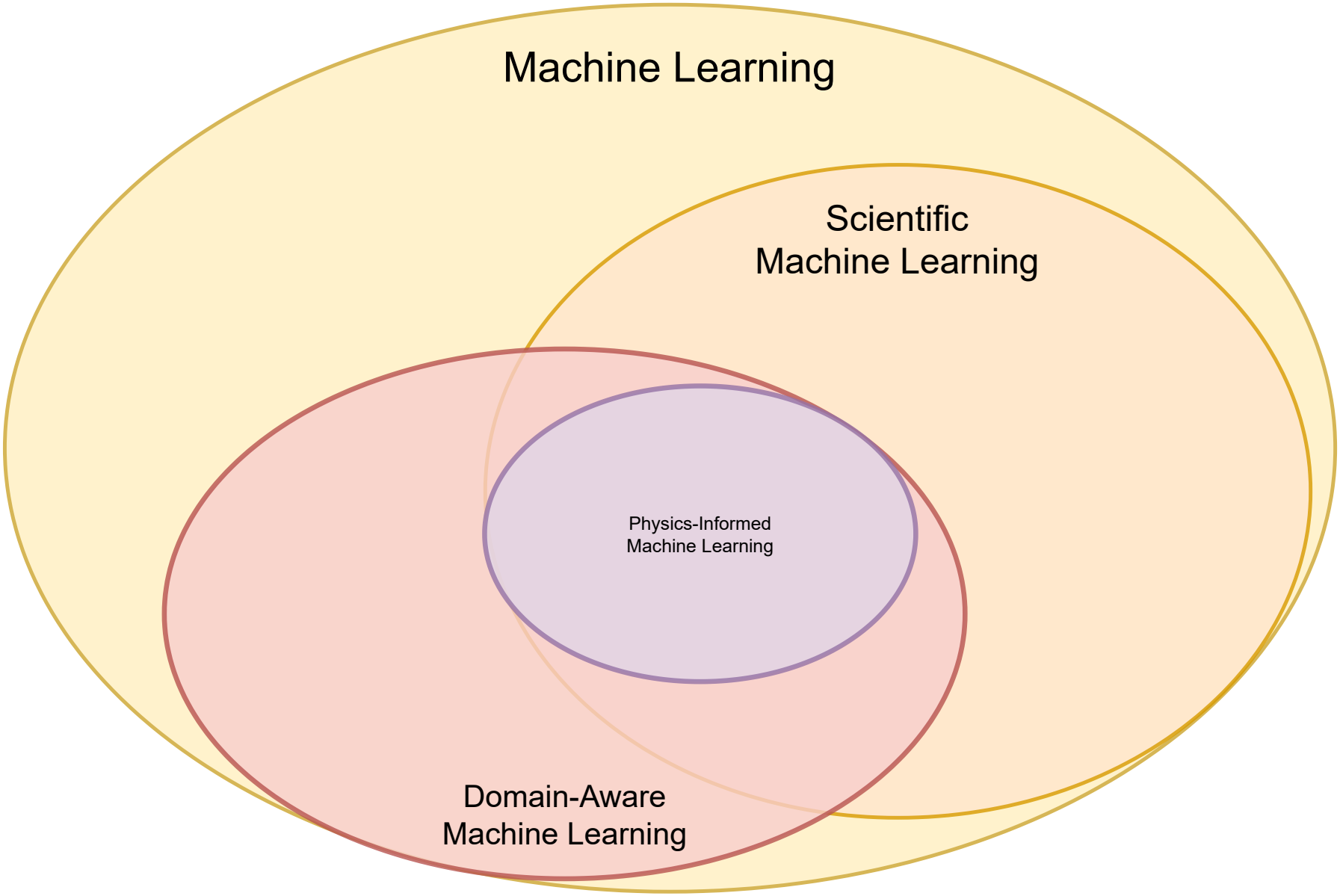
More generous definition, applies to

- Physics/natural sciences
- Engineering domains
- Domains with "soft" (non-formalized) knowledge (psychology, social sciences)
- Humanities (?)

Venn Diagram



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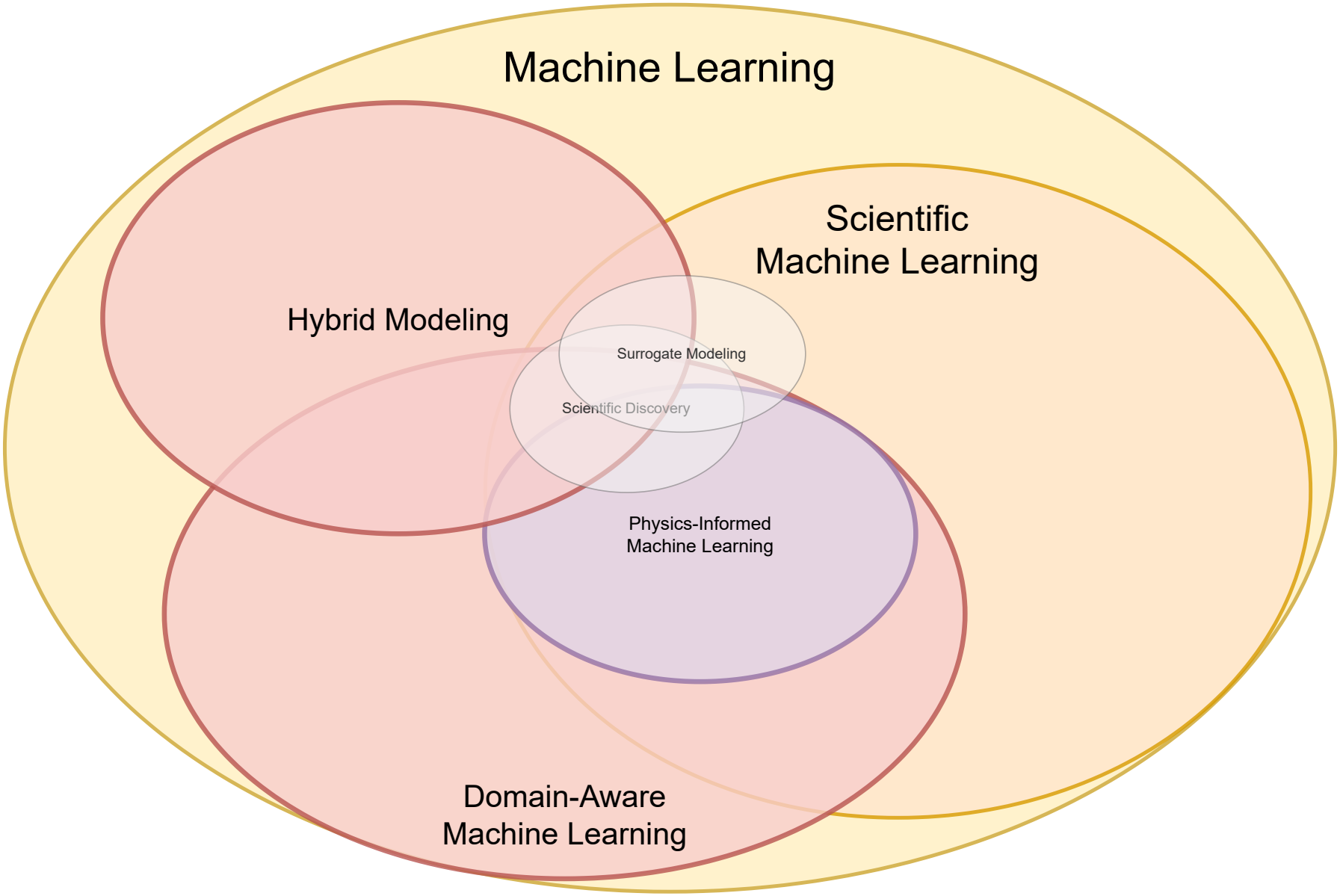

Motivation

Hybrid Modeling and Domain-Aware ML

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Why should we use Domain-Aware ML

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- instill other desirable properties into data-driven models

A Cost-Based Perspective on Machine Learning

Machine learning means trading between *at least* three objectives/costs:

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Each ML task has a Pareto front in this three-dimensional space.
Each concrete ML model is a point in this three-dimensional space.

A Cost-Based Perspective on Machine Learning (cont'd)

Domain-Aware Machine Learning

Domain-aware ML allows getting closer to the Pareto front (**knowledge is free!**) compared to purely data-driven ML.

- Ensure that ML model extrapolates consistently (*predictive performance*)

A Cost-Based Perspective on Machine Learning (cont'd)

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- Achieve same model accuracy with a smaller, domain-informed model (*model evaluation effort*)
- Achieve same model accuracy with less training data (*model creation effort)*
- Related to Green AI (?)

A Brief Intro to Machine Learning

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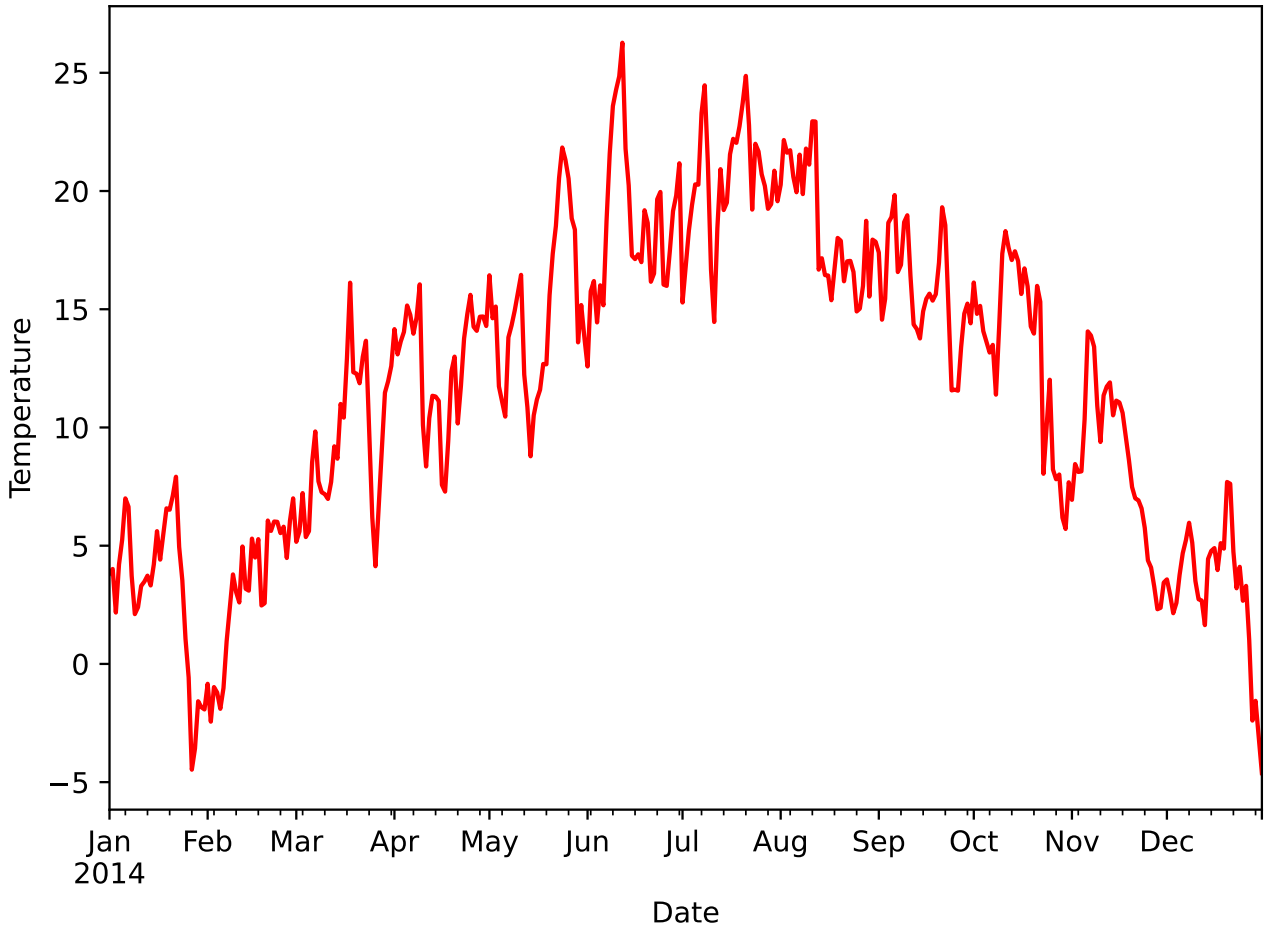

Temperature (y) recorded in Graz throughout 2014, shown as a function of time (x).
Data from `https://doi.org/10.5281/zenodo.3982670`.

A Brief Intro to ML

Given a dataset $\mathcal{D} = \{(x_i, y_i)\}$ of N pairs of features and targets, learn a function that "best" describes the relationship between x and y.

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$$f^\bullet \in \operatorname*{argmin}_{f \in \mathcal{F}} \frac{1}{|\mathcal{D}|} \sum_{(x_i, y_i) \in \mathcal{D}} \ell(y_i, f(x_i)) + \lambda R(f).$$

The Learning Problem and its Ingredients (cont'd)

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Data $\mathcal{D}$

- contains realizations of features x and targest y
- represents (hopefully) the relationship we want to learn

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Loss function $\ell \colon \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}$

- encapsulates our understanding of what amounts to a "good" function $f^\bullet$ that describes the relationship between x and y

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Model class $\mathcal{F}$

- restricted set of candidate functions f over which we search
- chosen to simplify the learning problem
- follows from selecting a certain ML algorithm/approach
- e.g., a regression tree of depth d yields piecewise constant functions with at most 2^d pieces
- e.g., a neural network of a given architecture yields functions within a class $\mathcal{F}$ defined by this architecture

The Learning Problem and its Ingredients (cont'd)

$$\theta^\bullet \in \underset{\theta \in \Theta}{\operatorname{argmin}} \frac{1}{|\mathcal{D}|} \sum_{(x_i, y_i) \in \mathcal{D}} \ell(y_i, f_\theta(x_i)) + \lambda R(f_\theta).$$

Model parameterization θ

- every function $f \in \mathcal{F}$ can be described by a parameter vector $\theta \in \Theta$
- e.g., a neural network is parameterized by the set of weights and biases
- e.g., a regression tree of depth d can be parameterized by the split positions and leaf weights; or by intervals and scaling terms $\rightarrow$ parameterization is not unique!
- choosing the parameterization can strongly influence the learning problem

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Regularization R

- represents our preference for certain functions within $\mathcal{F}$
- often encapsulates our preference for parsimonious models (Occam's razor)

The Learning Problem and its Ingredients (cont'd)

$$\hat{f} = \mathsf{A}_{\mathcal{F},\ell}(\mathcal{D}, R, \lambda, \beta, \dots) \approx f^\bullet \in \operatorname*{argmin}_{f \in \mathcal{F}} \frac{1}{|\mathcal{D}|} \dots$$

Optimization approach A

- is an algorithm that has as output a (parameterization $\hat{\theta}$) of a function $\hat{f}$
- strongly affects how close $\hat{f}$ is to $f^\bullet$
- is often tied to a given model class and/or loss (e.g., SGD for neural networks, C4.5 for decision trees, genetic algorithm for symbolic regression, etc.)

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Domain-Aware ML – Overview

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The Learning Problem and its Ingredients[1]

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How can domain knowledge improve these ingredients?

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Domain knowledge can affect data $\mathcal{D}$:

- selection (smart data vs. big data)
- augmentation
- preprocessing/feature engineering

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Domain knowledge can affect loss function ℓ or regularizer R:

- consistency with algebraic (PGNN) or differential (PINN) equations
- monotonicity/positivity
- encouraging operational goals aside from accuracy
  - privacy/fairness
  - invariance
  - disentanglement (β-VAE, etc.)
  - ...

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Domain knowledge can affect model class $\mathcal{F}$ or its parameterization θ:

- anything we know about the relationship in the data can help narrowing down $\mathcal{F}$/parameterize the model
- select candidate functions for symbolic regression
- parameterize $\mathcal{F}$ to achieve non-negativity/monotonicity
- choose $\mathcal{F}$ to achieve translation invariance/equivariance (e.g., CNNs)
- enforce conservation laws (e.g., softmax layer)

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Domain knowledge can affect the optimization approach A

- rely on inductive bias of the optimizer (e.g., gradient descent finds minimum-norm solutions in certain settings)
- constrained optimization (related to loss/regularization/model class)
- custom optimization algorithms
- (to be honest, this is not yet fully clear)